

UNIVERSITY OF BOTSWANA

DEPARTMENT OF MATHEMATICS

2014/2015 – EXAMINATIONS

COURSE NUMBER MAT111 DURATION 2 hrs DATE May 2015

TITLE OF PAPER INTRODUCTORY MATHEMATICS I

SUBJECT MATHEMATICS TITLE OF EXAMINATION BSc I

MORNING/AFTERNOON

INSTRUCTIONS:

- ANSWER ALL QUESTIONS IN SECTION A AND ANY TWO(2) QUESTIONS FROM SECTION B.
- ALL MARKS ARE INDICATED IN BRACKETS [].

NUMBER OF PAGES INCLUDING COVER PAGE: 5

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TO DO SO BY THE SUPERVISOR.

SECTION A

Answer All Questions: Each Question carries 10 Marks

Question 1

(a) Factorize $x^3 - \frac{8}{27}$. [3]

(b) Evaluate $(-64)^{-\frac{2}{3}}$ without using a calculator. [3]

(c) Simplify $\frac{y^2 + 4y - 12}{3y - 6} \cdot \frac{12y + 24}{y^2 - 36}$. [4]

Question 2

(a) Simplify the fraction

$$\frac{4(2^{n-1}) - 3(2^{n+1})}{2^{n+1} - 2^n}.$$

[5]

(b) Find the point of intersection of the graphs of $2x - 3y = -2$
and $4x - y = 6$. [5]

Question 3

(a) Solve the inequality

$$|1 - \frac{x}{2}| < \frac{1}{2}.$$

[5]

(b) Let $f(x) = \frac{1}{x}$, $g(x) = \sqrt{2x+4}$. Determine the function
 $(f \circ g)(x)$ and find its domain.

[5]

Question 4

- (a) Let $f(x) = 4x^3 - 3x^2 + 9x - 2$. Show that $x + 1$ is a factor of $f(x)$. Solve the equation $4x^3 - 3x^2 + 9x - 2 = 0$. [7]
- (b) Write $z = \frac{1}{-1 + 2i}$ in the form $a + bi$ with a and b real. [3]

Question 5

- (a) Determine the values of x that satisfy the equation:
 $\ln(x - 1) + \ln(x + 3) = 3 \ln 2$. [6]
- (b) Solve: $4(3^x) = 7$. [4]

Question 6

- (a) Given that $\pi < \theta < \frac{3\pi}{2}$ and $\cos \theta = \frac{3}{5}$, find $\tan \theta$. [5]
- (b) Evaluate $\cot \frac{4\pi}{3}$ without using a calculator. [5]

SECTION B

Answer TWO(2) Questions OUT of FOUR(4):
Each Question carries 20 Marks

Question 7

- (a) Solve the inequality $\frac{2x-7}{x-5} \leq 3$. [7]
- (b) Find the center and radius of the circle $x^2 - 2x + y^2 + 6y + 6 = 0$. [6]
- (c) Given that $f(x) = \sqrt{x-1}$ and $g(x) = \sqrt{9-x^2}$, find $(f/g)(x)$.
Then find the domain of f/g . [7]

Question 8

- (a) Let $f(x) = \frac{3}{x-2} - \frac{3}{x+2}$.
- (i) Show that $f(x)$ is an even function,
- (ii) Determine which one of the two points $y_1 = \frac{12}{5}$ and $y_2 = -1$ is in the range of f .
(Hint: Express $\frac{3}{x-2} - \frac{3}{x+2}$ as a single fraction.) [8]
- (b) Show that the function $f : \mathbb{R} \rightarrow (0, \infty)$ given by $f(x) = e^{2x-1}$ is bijective and find $f^{-1}(x)$. [6]
- (c) Sketch the graph of $f(x) = \frac{x}{x^2 + x - 2}$. [6]

Question 9

- (a) One solution to the equation

$$x^4 + 3x^3 - 5x^2 - 21x + 22 = 0$$

is the complex number $x = -3 + \sqrt{2}i$. Determine all the solutions to this equation. [8]

- (b) Solve the equation $3x^2 + 2x - 4 = 0$ by completing the squares. [6]

- (c) Solve the equation $e^{2x} - 4e^x - 5 = 0$. [6]

Question 10

- (a) Prove the identity

$$\sec^2 x - \tan x \sec x = \frac{1}{1 + \sin x}.$$

[7]

- (b) Solve $3 \sec^2 x - 4 = 0$ for $0 \leq x \leq 2\pi$. [6]

- (c) Show that $\cos \frac{5\pi}{12} = \frac{\sqrt{2}}{4}(\sqrt{3} - 1)$. [7]

